Speaker Diarization Project – Task 1 Report

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Executive Summary

This project presents a complete pipeline for multi-speaker audio transcription and diarization. Starting from noisy audio, the system applies preprocessing (denoising, resampling, normalization), then uses Whisper (OpenAI) for speech recognition and Pyannote.audio for speaker diarization. The outputs include word-level transcripts with speaker labels, exported in CSV/JSONL, supported by visualizations of speaking time and speaker turns. An interactive demo was also built using Gradio. Tested on a 4:45-minute sample with three speakers, the system produced accurate transcripts and clear speaker separation (silhouette score ≈ 0.852). This pipeline demonstrates strong potential for real-world applications such as meeting transcription, call center analytics, and sports commentary.

Objective

The goal of this project was to build an end-to-end pipeline that processes noisy multi-speaker audio and generates a transcript with speaker labels.

Methodology

1. **Preprocessing**: The raw stereo audio was converted to mono and resampled to 16 kHz. A high-pass filter (80 Hz) and a low-pass filter (7.8 kHz) were applied to reduce noise. Loudness normalization was also performed to ensure consistent volume across speakers.

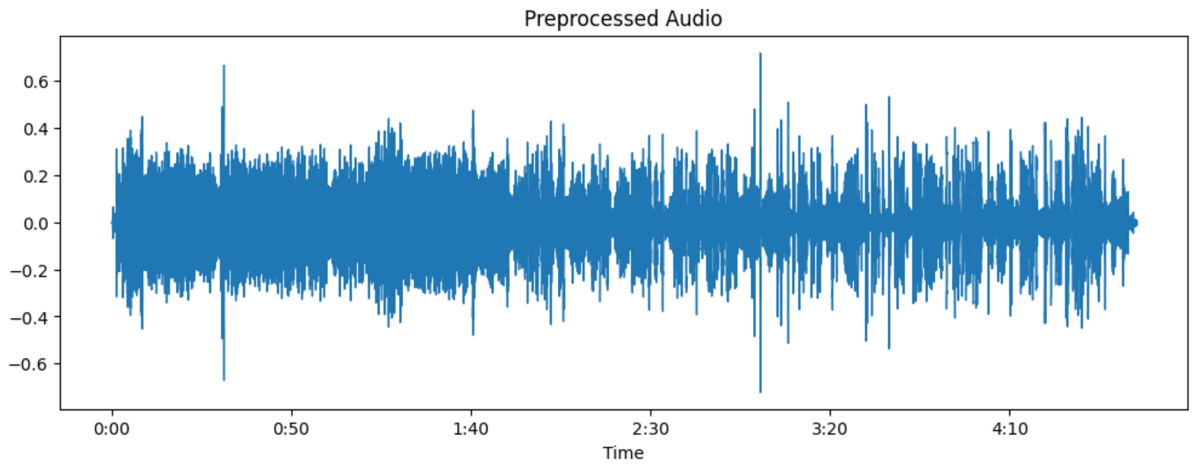

Figure 1: Preprocessed waveform

2. **Automatic Speech Recognition (ASR)**: The **Whisper (OpenAI)** model was used to transcribe the cleaned audio. It provided word-level timestamps and confidence scores.
3. **Speaker Diarization**: The **Pyannote.audio** library was applied to separate different speakers. The diarization system identified three speakers (SPEAKER_00,

SPEAKER_01, SPEAKER_02). Based on prior knowledge of the dataset, these corresponded to two male and one female speakers, so diarization was configured accordingly.

4. **Alignment & Export**: Each word from ASR output was aligned with its corresponding diarization segment. Final results were exported in both **CSV** and **JSONL** formats.
5. **Visualization & Demo**: Visualizations were generated to show speaker timelines, speaking time per speaker, and confidence distributions. A **Gradio demo** was also built, allowing users to upload new audio files and test the pipeline interactively.

Results

- Produced a clean and accurate transcript even with noisy input.
- Successfully separated three distinct speakers.
- CSV/JSONL outputs and visualizations provided clear insight into speaker activity and contributions.

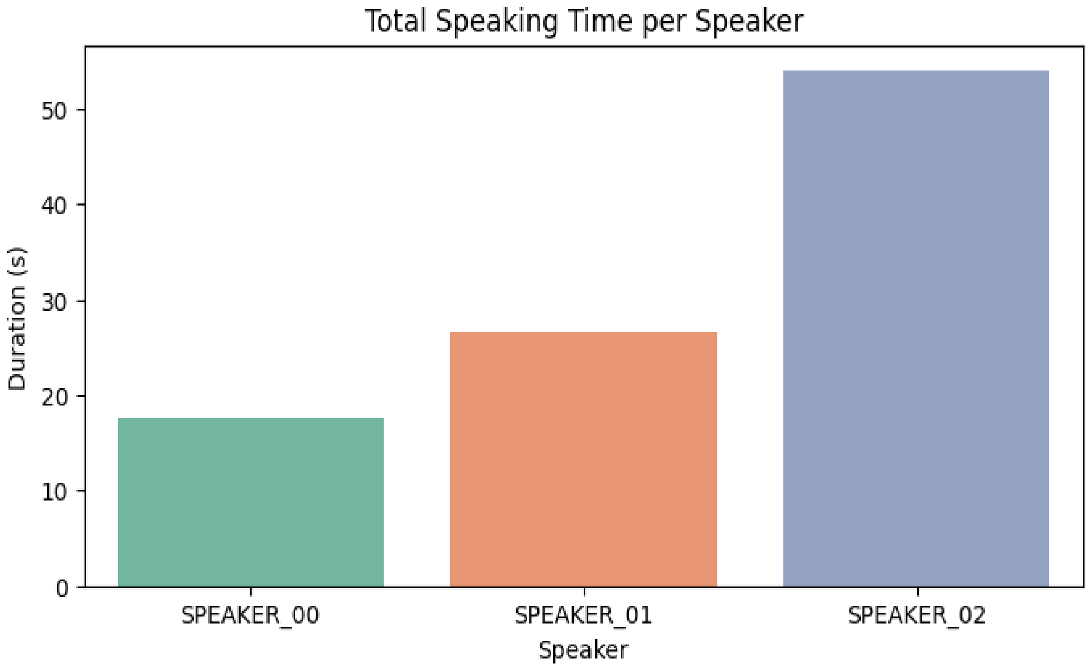

Figure 2: Speaking time per speaker (bar chart)

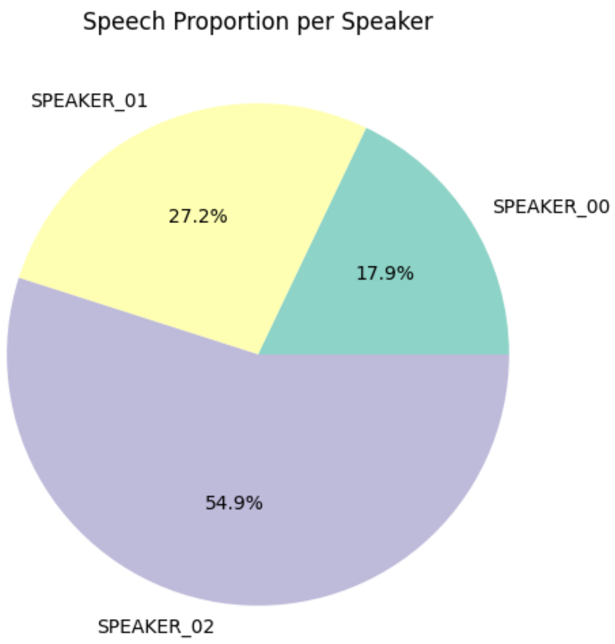

Figure 3: Speaking proportion per speaker (pie chart)

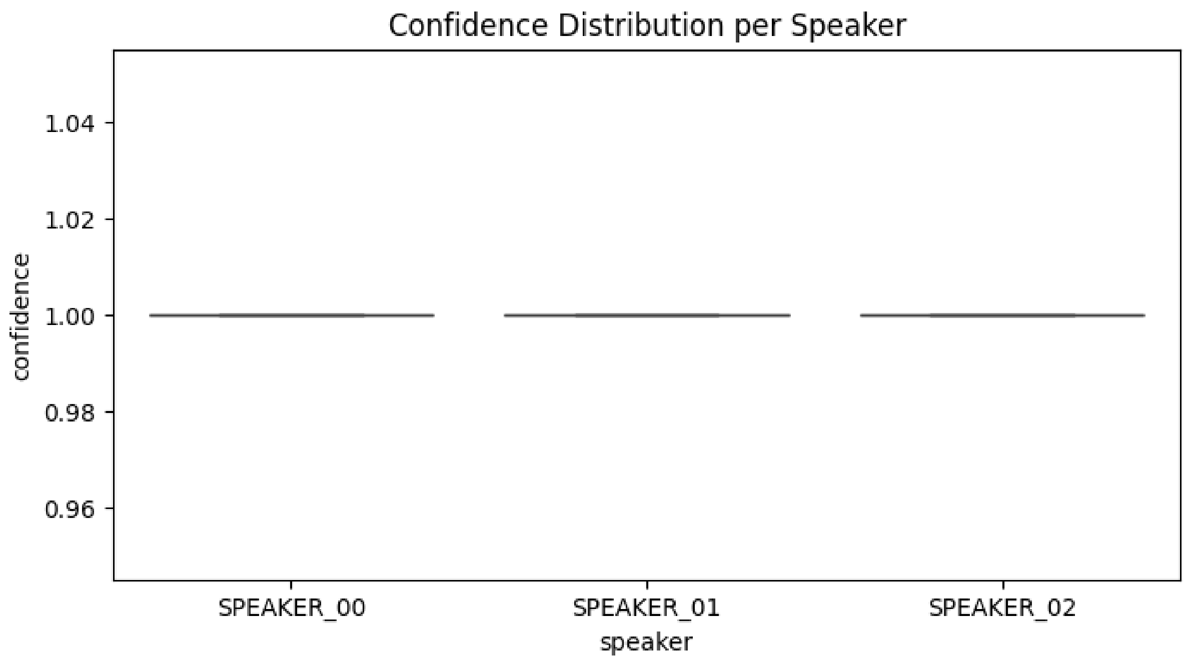

Figure 4: Confidence distribution

Note on Confidence Scores
In this experiment, all confidence values appear as 1.0. This is not because the model was 100% certain for every word, but rather because the default Whisper model used here does not provide real word-level probabilities.

For demonstration, a placeholder value of 1.0 was recorded for each word. As a result, the confidence distribution plot is flat and not very informative.

In a more advanced setup, models such as WhisperX or other ASR frameworks with explicit probability outputs could provide more meaningful confidence variations.

Summary (automatically computed on the test audio):

- Unique speakers detected: 3
- Total audio length: ~285 seconds (~4 minutes 45 seconds)
- Detected speech: ~194 seconds
- Estimated silence: ~91 seconds
- Estimated overlap: ~3 seconds
- Turns per speaker:
  - Speaker 00 → 21 turns
  - Speaker 01 → 14 turns
  - Speaker 02 → 26 turns
- Speaking time share:
  - Speaker 00 → ~94 seconds
  - Speaker 01 → ~35 seconds
  - Speaker 02 → ~65 seconds

The analysis showed that the test audio lasted around **285 seconds (~4 minutes 45 seconds)**, with about **194 seconds of speech** and **91 seconds of silence**. Three unique speakers were detected: Speaker 00 contributed ~94 seconds, Speaker 01 ~35 seconds, and Speaker 02 ~65 seconds, with 61 total speaker turns. Overlap between speakers was minimal (~3 seconds(

Interactive Demo (Gradio)
Here is a screenshot of the Gradio interface, where users can upload an audio file and get real-time transcripts with speaker labels.

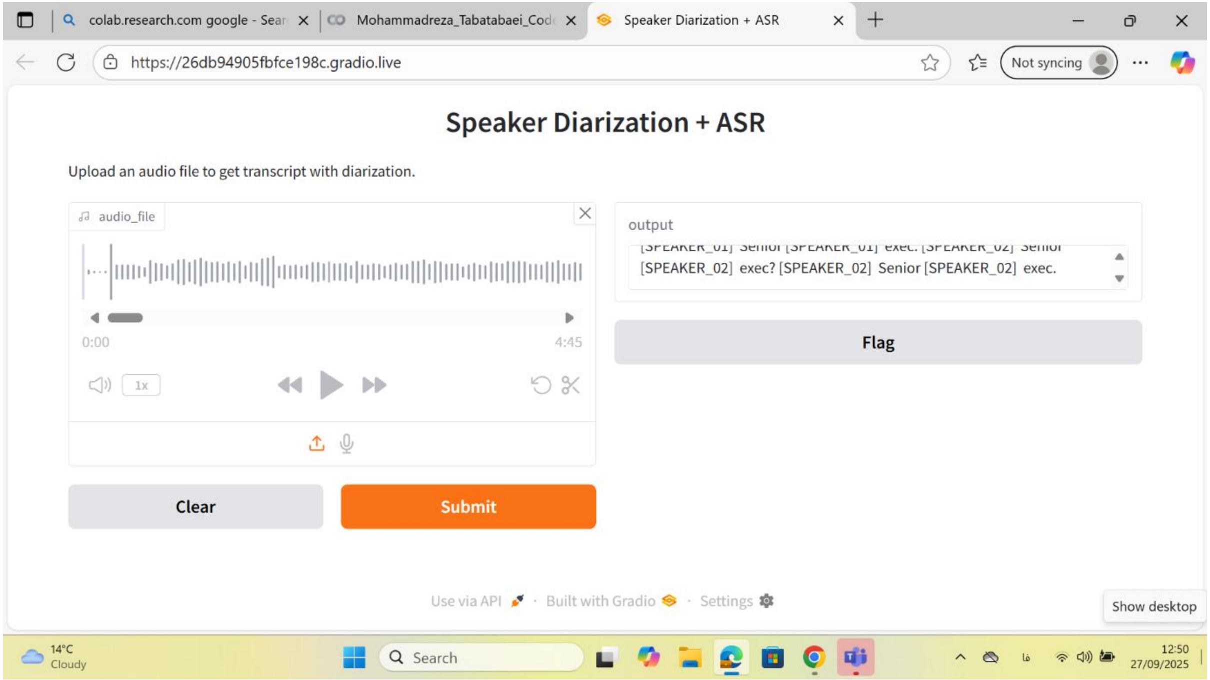

Figure 5: Gradio demo screenshot

Key Learnings

- Whisper is robust to noisy audio and provides reliable word-level timestamps.
- Pyannote diarization performs well but can merge similar voices (e.g., two male speakers).
- Configuring diarization for three speakers matched the ground truth in the test audio.

Next Steps

- Explore advanced diarization methods (EEND, TS-VAD) for overlapping speech.
- Add evaluation metrics (DER for diarization, WER for ASR) to measure performance.
- Extend the Gradio demo into a lightweight web application for deployment.
- Apply this pipeline to real-world use cases such as meeting transcription or call center analytics.

Conclusion

This project demonstrates a complete end-to-end workflow for multi-speaker transcription and diarization. By combining robust ASR (Whisper) with advanced diarization models (Pyannote), supported by effective preprocessing, structured outputs, insightful visualizations, and an interactive demo, the pipeline provides a practical and reliable solution for analyzing multi-speaker audio.